

A Further Discussion of Significance Testing, Power Analysis, and Post-Hoc Analysis

Null hypothesis significance testing: In this paper, we work within the framework of null hypothesis significance testing (NHST). NHST is not free from problems, in that certain systematic processes within the practice of scientific research and publishing can undermine its advantages, many of which have been explored in the literature (Gelman and Loken, 2013; Ioannidis, 2019; McShane et al., 2019). Nevertheless, it would be premature to discard the entire paradigm, and we believe there is still some value in considering power within NHST for several reasons.

First, despite its flaws, NHST remains a commonly used experimental framework in NLP research. Whether implicit or explicit, most experimental comparisons in the NLP literature have the structure of an experiment in the NHST framework, where having equivalent performance to an existing baseline is treated as a null hypothesis and the new model is argued to be significantly better (the typical case) or significantly worse (far rarer). But, whereas many fields that run experiments have standardized procedures for assessing statistical significance, NLP papers vary as to how formally they use a hypothesis testing framework to evaluate their results (Berg-Kirkpatrick et al., 2012; van der Lee et al., 2019; Azer et al., 2020).

Second, when done properly, NHST does provide a convenient way of summarizing results. Improvements in overall methodology, such as sharing code and data, sensitivity analyses, greater interest in null findings, and even pre-registration can vastly improve the validity of this paradigm, and we are seeing adoption of some of these practices within NLP.

Finally, there is also a great need for additional clarity with respect to precisely what claims are being made by NLP papers. In this work, we are primarily focused on claims made about trained models (i.e. in testing whether one particular instantiation of a model is significantly better than a particular instantiation of another model). It is, of course, also important to consider broader claims that might be made, such as about expected performance or computational budget (Dodge et al., 2019; Schwartz et al., 2019), and everything we have to say can be extended to incorporate such considerations. For the purpose of clarity, how-

ever, we restrict ourselves to the simplest sort of statistical claim.

Power and power analyses: The probability that a statistical test will reject the null hypothesis in an experiment is a function of several parameters, some of which are typically known or controllable, such as the sample size and significance threshold, and some of which are unknown, such as the details about exactly how models differ. Power tells us what this probability would be, if we knew the true values for these unknown parameters. Conditional on a particular difference existing (e.g. an expected difference in accuracy between two models for a particular data distribution), along with a statistical test, a significance threshold, power is the probability that the test will reject the null hypothesis and find the observed difference to be significant. In common statistical terminology, power is one minus the probability of false negatives in rejecting the null hypothesis or type II error.

While we will not, in general, know what the true power of an experiment is, by making reasonable assumptions, we can try to choose appropriate values for those parameters that we can control. By making assumptions about what we expect to observe, we can obtain estimates of how much power a test is likely to have, which may lead us to modify our experimental design, such as by increasing the sample size.

Importantly, proper experiment design requires specifying these parameters in advance of data collection, or otherwise using a valid stopping rule. One can *always* obtain a significant result by progressively collecting data until a significant result is found (“sampling to a foregone conclusion”), but this is not a valid procedure (Anscombe, 1954; Wagenmakers, 2007). Similarly, *post-hoc* power analysis, using estimates derived from the experiment itself, provides no additional information beyond a transformation of the observed p -value, and is thus not recommended (though see below).

Expanding on the algorithm in Figure 2, a simulation-based power analysis involves the following:

1. First, determine the statistical test, T , which will be used. For the example of comparing models depicted in Figure 1, we will use the binomial test to compare the systems (Dror et al., 2018).
2. Come up with a generative process which

could be used to generate data like that which we will collect. In this step, we need to make assumptions about the comparison of interest. Since the binomial test requires only the counts of how many people prefer each system, we need to specify a prior on generating those counts. For example, we might assume that 60% of people will prefer system B, so the generative process will be $c_B \sim \text{Binomial}(p = 0.6, n)$, where n is the total number of people to be sampled.

3. Choose a value of n for which we want to calculate power. Repeatedly (e.g., 10,000 times) draw many samples from our assumed generative process for that size of n .
4. For each simulated dataset of size n , run the chosen statistical test to check if difference between the observed counts is significant, and compute the proportion that are found to be significant. This is our estimate of power.

Note that more direct solutions for power analysis do exist for some settings, such as this one (see Appendix E.5 below).

Post-Hoc Power Analysis: Post-hoc power analysis is an issue when the true population effect has variance to it (O’Keefe, 2007; Hoening and Heisey, 2001; Gelman, 2019). In the case of NLP models, there are several perspectives on the comparisons which can lead to differences regarding how we perceive post-hoc power analysis: (1) we are comparing one model vs. another on a particular test set, the effect we see is the true population effect, post-hoc power analysis is okay because it is deterministic; (2) we are comparing one model vs. another on a data distribution from which the test and dev set are drawn, post-hoc power is not okay; (3) we are comparing one training algorithm vs. another (including variance from both training procedures and test/dev set draws), post-hoc power analysis is still not okay. We specifically look at the case of (2). While (3) is interesting on its own, this is not the typical comparison done (yet) in NLP research and thus we do not have enough information on reported training variance to investigate this thoroughly here. The case of (1) is also atypical as the authors of a study typically wish to draw inferences about how well a model does on the true data distribution (hence, why a dev and test set are used).

B Type-M and Type-S errors

Although the most obvious risk of using underpowered experiments is that there is a greater chance of failing to detect a true effect, there is an additional harm of using an underpowered design, which has emerged in light of the replication crisis in science. This can be most easily understood through the idea of Type-M and Type-S error (Gelman and Carlin, 2014).

Type-M error is the extent to which an observed difference exaggerates the true effect, conditional on a finding being significant. Type-S error is the probability that an observed difference has the opposite sign of the true difference, again conditional on a finding being significant. Even in a low-powered experiment, there is some probability of finding an effect to be significant; the lower the power, however, the more likely it is that the observed significant difference has the opposite sign of the true effect, and the larger the degree to which the magnitude of the observed effect will tend to exaggerate the true effect.

Intuitively, if power is low, this means that the sample size is small relative to the effect size. As such, the difference will *only* be significant if an atypically large effect is observed. Assuming the use of a two-sided test, many of these significant findings will also have the wrong sign, as they will be nearly as likely to fall on either side of zero for a symmetric distribution.

Type-M and Type-S error rates can be estimated using the exact same process for power analysis as described in Figure 2. To do so, we need only augment the algorithm with these two additional steps:

$$3. \text{ Type-S error} \approx \sum_{i: p_i \leq \alpha} \frac{\mathbb{I}[\text{sign}(e_i) \neq \text{sign}(e^*)]}{|j: p_j \leq \alpha|}$$

$$4. \text{ Type-M error} \approx \sum_{i: p_i \leq \alpha} \frac{\text{abs}(e_i)/\text{abs}(e^*)}{|j: p_j \leq \alpha|}$$

Figures 7 and 8 show scenarios for comparing classifiers on accuracy, corresponding to Figure 3 in the main text, but showing expected Type-M and Type-S error instead of power. As can be seen, Type-M and Type-S error increase with smaller sample sizes, smaller differences between models, and lower agreement rates, all corresponding to lower power.

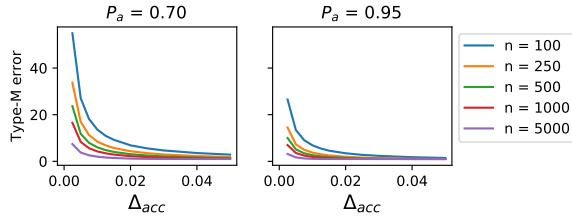


Figure 7: Type-M error (the factor by which observed significant effects are likely to exaggerate the true effect) for comparing classifiers on accuracy increases with smaller test sets (n), smaller differences between models (Δ_{acc}), and smaller agreement rates (P_a). Severe exaggerations of differences between models are likely with underpowered designs.

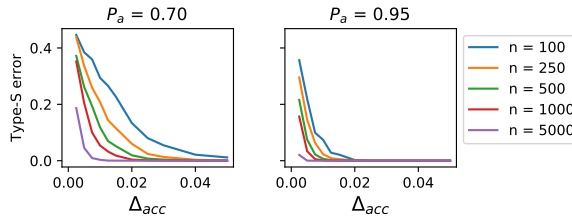


Figure 8: Type-S error (the probability that significant differences observed between models will have the opposite sign of the true difference) for comparing classifiers increases with smaller test sets (n), smaller differences between models (Δ_{acc}), and smaller agreement rates (P_a). Sign errors become reasonably likely with underpowered experiments.

C Numerical Example of a McNemar’s Test Simulation

To provide a concrete example of comparing classifiers on accuracy, imagine that a test set for a benchmark task has 500 instances. Based on prior knowledge (see main paper), we might assume that our proposed model will achieve, at most, an absolute improvement of 2 percentage points over the state of the art ($\Delta_{acc} = 0.02$), and that the models are likely to agree on 90% of examples ($P_a = 0.9$). We can convert these assumptions into a distribution over outcomes which will define our generative process. In particular, for a random unseen instance, these assumptions imply that there is a 10% chance of a disagreement; the probability that our model is correct and the old model is incorrect is therefore 6%, and the opposite outcome has a probability of 4% (giving us the assumed net difference of 2%). Note that, because McNemar’s test does not consider the on-diagonal elements, it is not necessary that we explicitly define the baseline accuracy. Thus, a valid probability distribution

	M1 correct	M1 incorrect
M2 correct	0.6	0.06
M2 incorrect	0.04	0.3

Table 4: A possible distribution corresponding to the case where models M1 and M2 will agree on 90% of examples (P_a) and M2 achieves a 2% improvement over M1 (Δ_{acc}). Note that the on-diagonal terms here will be dictated by the accuracy of M1 (or equivalently, by M2), but for our purposes, only need to be non-negative and sum to P_a for the sake of McNemar’s test, which only looks at the off-diagonal elements.

for use in this simulations could be that shown in Table 4.

By drawing many samples from this distribution of size $n = 500$ and computing a p -value using McNemar’s test for each, we obtain an estimate that the power of this test is approximately 0.25 for a significance threshold of $\alpha = 0.05$, which is severely underpowered. This would also imply a Type-M error factor of 1.9; we would expect that a typical experiment that found the observed difference between models to be significant would exaggerate the true difference of 0.02 by a factor of 1.9, producing observed significant differences between models on the order of 0.04, on average. (See supplementary notebooks for calculations and interactive demonstration). As such, we conclude that this test set is too small to be able to reliably evaluate whether or not our model is significantly different from the state of the art, and should distrust any observed differences that are significant, unless we have poorly estimated the relevant parameters.

By contrast, if the test set contained 2000 examples, we would estimate the test to have nearly 80% power, with a Type-M factor of only 1.1, and would feel comfortable proceeding with and reporting on this evaluation. Similarly, if we had reason to think that our model represented a game-changing advance, and would achieve an improvement of 4 percentage points, or if we had reason to believe that the models would agree on 97.5% of examples, then we would have the power to evaluate this, even with only 500 examples.

D SQuAD 2.0 Analysis and Results

From the authors of SQuAD 2.0, we obtained pairwise agreement statistics on the SQuAD 2.0 development and test sets for all models that were submitted to the SQuAD 2.0 leaderboard and had